

One Dimensional Analysis of a Viscous Reacting Shock Wave

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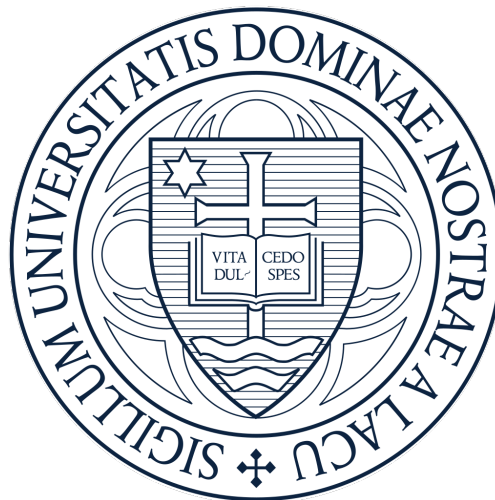
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Abstract

This study predicts the dissociation of molecular nitrogen across a shock wave. The shock wave profile is modeled using the reactive Navier–Stokes equations, augmented by constitutive relations and Dalton’s law to describe the behavior of a reacting gas mixture. Chemical kinetics are employed to construct a system of differential equations governing the evolution of both reactive and inert shock structures. In addition, Rankine-Hugoniot jump conditions are used to define integration boundary conditions. Numerical integration is carried out using a fourth-order Runge–Kutta scheme to solve this coupled system. The results include detailed profiles of the velocity, temperature and concentrations of the species, allowing a comparative analysis between the reactive and inert cases. The results show that the temperature increase as a result of the compressibility across the shock wave is substantial enough to cause dissociation of the nitrogen molecules. The dissociation of molecular nitrogen is not enough to significantly change the overall shock structure. Furthermore, the influence of shock strength on molecular dissociation is examined. As the shock speed increases, the resulting higher post-shock temperatures lead to more pronounced dissociation of molecular nitrogen.

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Nomenclature

$\underline{A}$	reactive shock coefficient matrix	λ	mean free path mm
a_j	collision frequency factor	M	Mach number
α	thermal diffusivity $\frac{\text{m}^2}{\text{s}}$	M_i	molar mass $\frac{\text{kg}}{\text{kmol}}$
$\vec{b}$	reactive shock constant vector	μ	dynamic viscosity Pa · s
β_j	temperature dependency exponent	n	number of moles mol
c_p	constant pressure specific heat $\frac{\text{kJ}}{\text{kg} \cdot \text{K}}$	ν	kinematic viscosity $\frac{\text{m}^2}{\text{s}}$
c_v	constant volume specific heat $\frac{\text{kJ}}{\text{kg} \cdot \text{K}}$	ν_{ij}	net stoichiometric coefficient
e	specific internal energy $\frac{\text{kJ}}{\text{kg}}$	ν'_{ij}	stoichiometric coefficient (products)
$\bar{e}$	molar internal energy $\frac{\text{kJ}}{\text{kmol}}$	ν''_{ij}	stoichiometric coefficient (reactants)
G_j^0	Gibbs free energy $\frac{\text{kJ}}{\text{kmol}}$	p	pressure Pa
h	specific enthalpy $\frac{\text{kJ}}{\text{kg}}$	Pr	Prandtl number
$\bar{h}$	molar enthalpy $\frac{\text{kJ}}{\text{kmol}}$	q	heat flux $\frac{\text{W}}{\text{m}^2}$
i	species index	R	specific gas constant $\frac{\text{kJ}}{\text{kmol} \cdot \text{K}}$
j	reaction index	$\bar{R}$	universal gas constant $\frac{\text{kJ}}{\text{kmol} \cdot \text{K}}$
j_i	mass diffusion flux $\frac{\text{kg}}{\text{m}^2 \cdot \text{s}}$	r_j	reaction rate $\frac{\text{kmol}}{\text{m}^3 \cdot \text{s}}$
J	number of reactions	T	temperature K
k	thermal conductivity $\frac{\text{W}}{\text{m} \cdot \text{K}}$	t	wave frame time s
k_j	Arrhenius rate $\left(\frac{\text{kmol}}{\text{m}^3}\right)^{(1-\sum_{j=1}^J \nu'_j)} \cdot \text{s}^{-1}$	τ	viscous shear stress Pa
K_c	equilibrium constant $\left(\frac{\text{kmol}}{\text{m}^3}\right)^{\sum \nu' - \sum \nu''}$	u	wave frame velocity $\frac{\text{m}}{\text{s}}$
K_p	pressure equilibrium constant	$\hat{t}$	lab frame time s
		$\hat{u}$	lab frame fluid velocity $\frac{\text{m}}{\text{s}}$
		U	wave speed $\frac{\text{m}}{\text{s}}$
		x	wave frame position m
		$\vec{Z}$	reactive shock state vector
		$\hat{x}$	lab frame position m
		y_i	mole fraction

1 Introduction

1.1 Research Objectives

This study examines a left-moving wave at Mach 2.5 analyzed in the wave reference frame. The goal is to analyze the effects of nitrogen dissociation on the shock's structure and thermodynamic properties. The wave speed can be visualized from either the laboratory frame or the wave reference frame. The laboratory frame holds a fixed observer as the shock moves from right to left, whereas the wave frame entails the observer traveling with the shock, as shown in Figure 1.

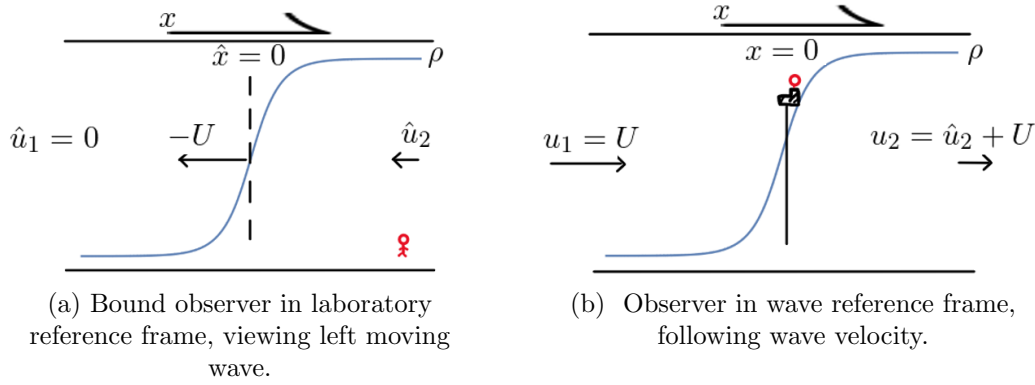


Figure 1: Fluid velocities in different reference frames

Figure 1 shows the pre and post-shock velocities $\hat{u}_1$ and $\hat{u}_2$ in the laboratory reference frame. Whereas, u_1 and u_2 are the unshocked and shocked fluid velocities in the wave reference frame. Furthermore, Figure 1 shows the transformations between the two reference frames, with the respective directional changes. The shock thickness is shown across the density profile, where ρ is the shock density.

The Navier-Stokes equations were used to exhibit the internal structure of a shock wave. A steady-state assumption was made to allow for the structure of the shockwave to be analyzed at a single moment in time. This focused the study on the initial structure of the shock wave rather than on how it propagated over time. After assuming a steady state, the conservation equations were manipulated into a system of differential equations. Utilizing

the pre-shock state as a boundary condition, the differential equations were solved to model the structure of the shock. Once the system model was solved, the rest of the flow properties were solved using constitutive models and simplified versions of the conservation equations. The mixture models were used to calculate the mass diffusion and reaction properties of a shock wave.

Next, chemical kinetics were employed to analyze the evolution of individual species that resulted from the sharp temperature increase across the shock wave. This analysis focused on the spatial development of the system, tracking how the concentrations of key species change as a function of distance through the shock structure. This information will further our understanding of shock wave(s) composition.

Finally, a FORTRAN script was written for both inert and reactive models to analyze the shock wave. The script utilizes the CHEMKIN chemical kinetics library CHEMKIN. The script defines a set of differential equations, as outlined in Section 4, then numerically integrates the coupled system from the post-shock solution to the pre-shock solution.

1.2 Literature Review

Here we review the literature of viscous shock waves. There are both theoretical and experimental studies that outline the structure of a shock wave. Using these studies as a foundation, we build our model. Early theoretical studies on the structure of shock waves were conducted by researchers such as Vincenti and Kruger [1], Liepmann and Roshko [2], and Zel'dovich [3]. While each of these works addresses a slightly different aspect of the problem, they are all grounded in the same fundamental principle: the shock structure must satisfy the Navier-Stokes equations at all points within its domain.

Vincenti and Kruger computed a model to analyze a shock structure of a right moving wave at $M = 1.7$. Their chosen medium was the non-inert gas argon. They chose this medium so that they could discover how close their findings matched with the experimental results of Talbot and Sherman from 1959 [4], who studied a (argon/non-inert gas)

medium. They discovered that their theoretical findings match the experimental results. Furthermore, Vincenti and Kruger outline the process of simple diatomic dissociation and equilibrium conditions. These conditions are used in calculating the post-shock conditions necessary for the integration procedure.

Thompson [5] outlines the importance of the shock structure. The inviscid assumption presupposes that the velocity and temperature gradients are infinite at the shock front, inducing infinite viscous stress and heat flux from the shock. To avoid the infinite viscous stress and heat flux, the extended *shock layer* reduces the velocity and temperature gradients. The internal shock layer is comparable to a boundary layer, in that both regions are characterized by large viscous stress and high heat flux bounded by external regions considered inviscid.

Liepmann and Roshko offer a more nuanced distinction between the two layers, while still enforcing the importance of the shock structure. The essential difference between the shock wave and the boundary layer lies in particle behavior. When a streamline enters a shock wave, it emerges downstream. Whereas when a streamline enters a boundary layer, it remains in a region of shear. Therefore, flowing through a shock wave, the particle transforms from one thermodynamic state to another. This distinction is widely accepted and mirrors the analyses found in [6, 7].

Zel'dovich expands this analysis of a shock wave structure into molecular reactions, analyzing the molecular dissociation and ionization of diatomic molecules across the wave. This research will set out the groundwork of the mathematical foundation of the work done by Zel'dovich. This study will dive into a simplified version of Zel'dovich's problem, analyzing the dissociation of molecular nitrogen across a shock wave.

Further experimental data show the existence of a viscous shock layer. In a study conducted by Sherman [8] to determine shock wave structures and relaxation periods, he found the experimental data followed the Navier-Stokes approximations while including losses due to viscosity and heat. Sherman's experiment utilized a shock holder, a type of shock-producing device that consists of a thin-walled circular cylinder or cone frustum immersed

into the uniform portion of a wind tunnel jet. Using this apparatus with the addition of temperature-sensitive resistance wires connected to a thermocouple, the system could analyze the temperature distribution across a shock. Figure 2 shows the dimensionless wire temperature, t_w , along the dimensionless wind tunnel length, y .

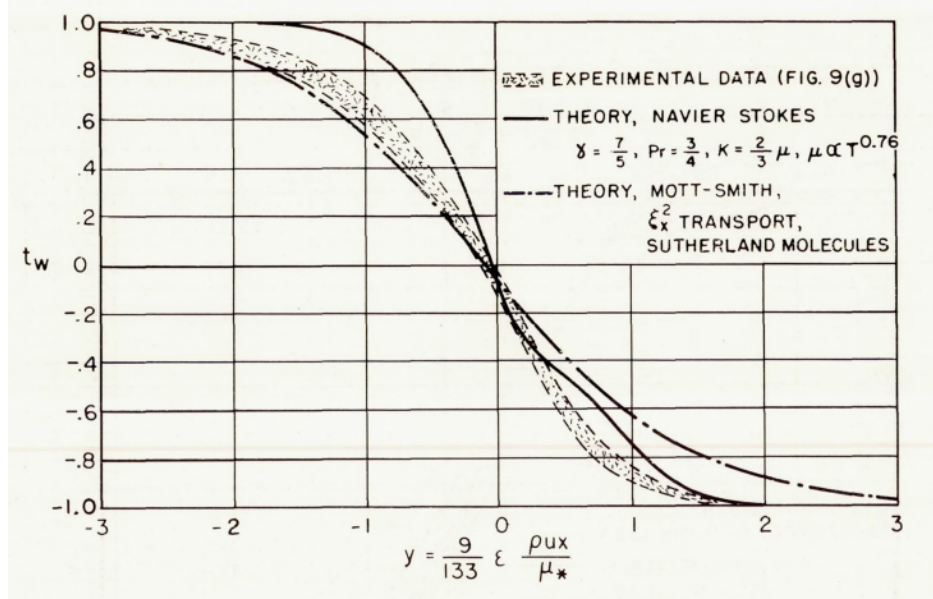


Figure 2: Comparison of various theories for a monatomic gas (helium). $M = 1.61$, Sherman [8].

In as recent as the past five years, studies have been analyzing the diffusivity in the shock structure and heat conduction. In 2021, Reddy [9], analyzed the heat conduction across the shock wave. The study utilized shock structure as a benchmark to evaluate the accuracy of different hydrodynamic models. By analyzing detailed profiles of density, temperature, and entropy production across the shock, the authors assessed how well the models capture the physical behavior of monatomic gases over a wide range of Mach numbers.

Additionally, in a study by Singh [10], a similar approach was used to analyze the shock wave structure as will be conducted later in this report. This study investigates the internal structure of plane shock waves in a two-phase gas-solid particle medium by solving the Navier-Stokes equations with thermodynamically consistent constitutive relations. By computing shock-front thickness, the analysis reveals that viscosity and particle concentra-

tion significantly influence shock width, with viscosity having a greater impact than heat conduction across various flow conditions.

2 Governing Equations

2.1 Mixture Model

To analyze the reactions occurring within a shock wave, a mixture model must be established. The Dalton model [11] is used. We will adopt the following rules for mixture properties:

$$p = \sum_{i=1}^2 p_i, \quad 1 = \sum_{i=1}^2 Y_i, \quad \rho = \sum_{i=1}^2 \rho_i = \sum_{i=1}^2 \frac{\bar{\rho}_i}{M_i}, \quad e = \sum_{i=1}^2 Y_i e_i, \quad (1)$$

$$0 = \sum_{i=1}^2 j_i, \quad h = \sum_{i=1}^2 Y_i h_i, \quad c_p = \sum_{i=1}^2 Y_i c_{pi}, \quad c_v = \sum_{i=1}^2 Y_i c_{vi}, \quad (2)$$

$$T = T_i. \quad V = V_i \quad (3)$$

The subscript i indicates that the property is a specific species property rather than the whole mixture. Here p is mixture pressure, Y_i is the species mass fraction, M_i is the molecular mass, e is the mixture internal energy h is the enthalpy, c_p is the mixture specific heat at constant pressure and c_v is the mixture specific heat at constant volume. Furthermore, p_i , ρ_i , $\bar{\rho}_i$, e_i , h_i , c_{pi} , c_{vi} , T_i , and V_i are species pressure, density, molar concentration, internal energy, enthalpy, constant pressure specific heat, constant volume specific heat, temperature, and volume.

2.2 Unsteady Conservative Form

The first step in analyzing the system is presenting the conservation equations and their different forms. The forms of all governing equations are standard and can be obtained

from [1, 12, 13]. To express the conservation of mass, linear momentum, and energy in conservative form and in the laboratory reference frame they are written as follows:

$$\frac{\partial \rho}{\partial \hat{t}} + \frac{\partial}{\partial \hat{x}}(\rho \hat{u}) = 0, \quad (4)$$

$$\frac{\partial}{\partial \hat{t}}(\rho \hat{u}) + \frac{\partial}{\partial \hat{x}}(\rho \hat{u}^2) + \frac{\partial}{\partial \hat{x}}(p - \tau) = 0, \quad (5)$$

$$\frac{\partial}{\partial \hat{t}} \left(\rho \left(e + \frac{\hat{u}^2}{2} \right) \right) + \frac{\partial}{\partial \hat{x}} \left(\rho \hat{u} \left(e + \frac{\hat{u}^2}{2} + \frac{p}{\rho} \right) + (\tau \hat{u} - q) \right) = 0. \quad (6)$$

Here, $\hat{x}$, $\hat{t}$ and $\hat{u}$ represent the position, time, and velocity of a particle in the laboratory reference frame. The variable q denotes the heat flux, while τ represents the shear stress. These forms are conservative because the quantities that are conserved are within the time derivative; i.e. mass, linear momenta, and energy. Expanding the heat flux, mass diffusion, and viscous shear stress as seen in Eq. (5) and Eq. (6), one obtains:

$$q = -k \frac{\partial T}{\partial \hat{x}} + \sum_{i=1}^2 j_i h_i, \quad j_i = -\rho \mathcal{D} \frac{\partial Y_i}{\partial \hat{x}}, \quad \tau = \frac{4}{3} \mu \frac{\partial \hat{u}}{\partial \hat{x}}. \quad (7)$$

The thermal conductivity of the fluid particle is k . Additionally, $\mathcal{D}$ is the mass diffusivity and μ is the dynamic viscosity. The thermal equation of state is defined using the mixture models in combination with the ideal gas law.

$$p_i = \bar{\rho}_i \bar{R} T = \rho \bar{R} T \frac{Y_i}{M_i} = \bar{R} T \frac{\rho_i}{M_i} = \rho_i R_i T. \quad (8)$$

Therefore the mixture pressure is:

$$p = \bar{R} T \sum_{i=1}^2 \bar{\rho}_i = \bar{R} T \sum_{i=1}^2 \frac{\rho Y_i}{M_i} = \bar{R} T \sum_{i=1}^2 \frac{\rho_i}{M_i} = \rho R T. \quad (9)$$

Here $\bar{R}$ is the universal gas constant and R is the specific gas constant of the mixture.

2.3 Unsteady Non-Conservative Form

The conservative equations can be transformed into their non-conservative counterparts by simple calculus. Using the product rule to expand, the mass, linear momenta, and energy equations can be simplified to the following:

$$\frac{\partial \rho}{\partial \hat{t}} + \hat{u} \frac{\partial \rho}{\partial \hat{x}} + \rho \frac{\partial \hat{u}}{\partial \hat{x}} = 0, \quad (10)$$

$$\rho \frac{\partial \hat{u}}{\partial \hat{t}} + \rho \hat{u} \frac{\partial \hat{u}}{\partial \hat{x}} + \frac{\partial}{\partial \hat{x}}(p - \tau) = 0, \quad (11)$$

$$\rho \left(\frac{\partial e}{\partial \hat{t}} + \hat{u} \frac{\partial e}{\partial \hat{x}} \right) + \frac{\partial q}{\partial \hat{x}} + \frac{\partial \hat{u}}{\partial \hat{x}}(p - \tau) = 0. \quad (12)$$

2.4 Galilean Transformation

The wave reference frame will be used for further analysis because it will simplify the subsequent calculations of thermodynamic properties. In this reference frame the observer moves along with the shock wave, effectively rendering it stationary. To convert between the laboratory reference frame and the steady wave reference frame, Galilean transformations are used:

$$x = \hat{x} + U\hat{t}, \quad t = \hat{t}. \quad (13)$$

Here x and t are the position and time within the wave reference frame, while $\hat{x}$ and $\hat{t}$ are the position and time in the laboratory reference frame. In this study, the wave moves to the left, therefore the distance traveled by the shock $U\hat{t}$ is added to the lab frame position to get into the wave reference frame. The position of the shock at $x = 0$ within the wave frame. Using the transformation in Eq. (13), we find how the derivatives are represented

under the coordinate transformation.

$$dx = \frac{\partial x}{\partial \hat{x}} d\hat{x} + \frac{\partial x}{\partial \hat{t}} d\hat{t} = d\hat{x} + U d\hat{t}, \quad (14)$$

$$dt = \frac{\partial t}{\partial \hat{x}} d\hat{x} + \frac{\partial t}{\partial \hat{t}} d\hat{t} = d\hat{t}. \quad (15)$$

By scaling dx by dt , we can derive the transformation for the velocities.

$$\frac{dx}{dt} = \frac{d\hat{x}}{d\hat{t}} + U, \quad (16)$$

$$u = \hat{u} + U. \quad (17)$$

Here we are taking the particle velocity in the wave frame $u = dx/dt$ and the particle velocity in the laboratory frame as $\hat{u} = d\hat{x}/d\hat{t}$. Furthermore, a generic dependent variable ψ can be defined such that it has a representation in the original space, $\psi(\hat{x}, \hat{t})$, and the transformed space, $\psi(x, t)$. These two points must map to the same space, such that:

$$\psi(\hat{x}, \hat{t}) = \psi(x, t). \quad (18)$$

Therefore the derivative of ψ can be expressed as the following:

$$d\psi = \left. \frac{\partial \psi}{\partial \hat{x}} \right|_{\hat{t}} d\hat{x} + \left. \frac{\partial \psi}{\partial \hat{t}} \right|_{\hat{x}} d\hat{t} = \left. \frac{\partial \psi}{\partial x} \right|_t dx + \left. \frac{\partial \psi}{\partial t} \right|_x dt. \quad (19)$$

Using the expressions of the wave frame derivatives found in Eq. (14) and Eq. (15), the expression can be simplified to:

$$d\psi = \left. \frac{\partial\psi}{\partial\hat{x}} \right|_{\hat{t}} d\hat{x} + \left. \frac{\partial\psi}{\partial\hat{t}} \right|_{\hat{x}} d\hat{t} = \left. \frac{\partial\psi}{\partial x} \right|_t dx + \left. \frac{\partial\psi}{\partial t} \right|_x dt, \quad (20)$$

$$= \left. \frac{\partial\psi}{\partial x} \right|_t (d\hat{x} + U d\hat{t}) + \left. \frac{\partial\psi}{\partial t} \right|_x d\hat{t}, \quad (21)$$

$$= \left. \frac{\partial\psi}{\partial x} \right|_t dx + \left(\left. \frac{\partial\psi}{\partial t} \right|_x + U \left. \frac{\partial\psi}{\partial x} \right|_t \right) d\hat{t}. \quad (22)$$

Now we consider Eq. (22) for constant $\hat{x}$, causing $d\hat{x} = 0$, and we divide by $d\hat{t}$.

$$\left. \frac{\partial\psi}{\partial\hat{t}} \right|_{\hat{x}} = \left. \frac{\partial\psi}{\partial t} \right|_x + U \left. \frac{\partial\psi}{\partial x} \right|_t. \quad (23)$$

Simplifying further the partial with respect to $\hat{t}$ becomes

$$\left. \frac{\partial}{\partial\hat{t}} \right|_{\hat{x}} = \left. \frac{\partial}{\partial t} \right|_x + U \left. \frac{\partial}{\partial x} \right|_t. \quad (24)$$

Furthermore, to analyze the steady state solution of a reaction, for constant t , $dt = 0$, therefore Eq. (22) reduces to:

$$\left. \frac{\partial}{\partial\hat{x}} \right|_{\hat{t}} = \left. \frac{\partial}{\partial x} \right|_t. \quad (25)$$

Eq. (25) is ultimately the transformation that can be used to redefine the governing equations from the laboratory frame to the wave reference frame. The result is invariant, meaning changing between the reference frames does not change the governing equations.

2.5 Steady Conservative Form

To convert the governing equations from unsteady to steady, time must remain constant. Therefore, all partial derivatives with respect to the independent variable t go to 0. Furthermore, all partial derivatives with respect to x become total derivatives, as described:

$$\frac{d}{dx}(\rho u) = 0, \quad (26)$$

$$\frac{d}{dx}(\rho u^2 + p - \tau) = 0, \quad (27)$$

$$\frac{d}{dx} \left(\rho u \left(e + \frac{u^2}{2} + \frac{p}{\rho} \right) + \tau u - q \right) = 0. \quad (28)$$

2.6 Steady Non-Conservative Form

Similarly to *steady conservative* forms of the governing equations, to transform the *unsteady non-conservative* governing equations into their steady counterparts, the time is taken to be constant. The system can therefore be written as,

$$u \frac{d\rho}{dx} + \rho \frac{du}{dx} = 0, \quad (29)$$

$$\rho u \frac{du}{dx} + \frac{d}{dx}(p - \tau) = 0, \quad (30)$$

$$\rho u \frac{de}{dx} + \frac{dq}{dx} + \frac{du}{dx}(p - \tau) = 0, \quad (31)$$

$$(32)$$

3 Chemical Kinetics

Chemical Kinetics describe the time-evolution of systems; typically describing the path to an equilibrium state. For the N and N₂ system, one dimensional kinetics can be used to describe the spatial excitation of the reaction. The evolution of chemical species in the wave frame is given by,

$$\frac{\partial}{\partial t}(\rho Y_i) + \frac{\partial}{\partial x}(\rho Y_i u + j_i) = M_i \dot{\omega}_i. \quad (33)$$

Here j_i is the mass diffusion flux of species i , and $\dot{\omega}_i$ is the species molar production rate. For the steady state reaction, ∂t goes to 0, leaving the reaction as:

$$\frac{d}{dx} (\rho Y_i u + j_i) = M_i \dot{\omega}_i. \quad (34)$$

Eq. (34) shows the evolution of chemical species, based off of the molar production rate and mass diffusion. From chemical kinetics, the species molar production rate $\dot{\omega}_i$ can be further expanded as,

$$\dot{\omega}_i = \sum_{j=1}^2 \nu_{ij} r_j. \quad (35)$$

Here r_j is given by the law of mass action, and ν_{ij} is the stoichiometric coefficient for each species, per reaction. The law of mass action is then expanded as,

$$r_j = k_j \prod_{i=1}^2 \bar{\rho}_i^{\nu'_{ij}} \left(1 - \frac{1}{K_{c,j}} \prod_{i=1}^2 \bar{\rho}_i^{\nu_{ij}} \right). \quad (36)$$

The reaction rate, k_j , given by the Arrhenius kinetics rule, $\bar{\rho}$ is the molar concentration and $K_{c,j}$ is the equilibrium constant.

$$k_j = a_j T^{\beta_j} \exp \left(\frac{-\bar{\mathcal{E}}_j}{RT} \right). \quad (37)$$

Here a_j is the collision frequency factor, β is a dimensionless parameter whose value is set by experiments, and $\bar{\mathcal{E}}_j$ is the activation energy needed for the reaction to occur. The equilibrium constant $K_{c,j}$ is defined as

$$K_{c,j} = \left(\frac{p_1}{RT} \right)^{\sum_{i=1}^2 \nu_{ij}} \exp \left(\frac{-\Delta G_j^0}{RT} \right). \quad (38)$$

The reference pressure is defined as p_1 , and $-\Delta G_j^0$ is the standard change in Gibbs free energy across the reactions. For this reaction the equilibrium states are calculated by defining the phase plane.

We will use chemical kinetics to calculate the final species concentration of the reaction. The final concentration is necessary to calculate the boundary conditions of integration. For our purpose we will need the equilibrium constant K_p , which is the equilibrium constant in terms of partial pressures.

$$K_p = \prod_{i=1}^2 \left(\frac{p_i}{p_1} \right)^{\nu_i} \quad (39)$$

If K_p is small, then it means that the reaction favors the reactants rather than the products. Once K_p is defined, we can solve for the final concentrations, from another definition of K_p :

$$\ln K_p = \ln \left(\frac{y_N^2 p}{y_{N_2} p_1} \right). \quad (40)$$

Here y_i is the mole fraction of the mixture. Using the definition of the mole fraction we can derive a single equation for the final moles of molecular N_2 .

$$K_p = \left(\frac{2(n_{N_2}|_{t=0} - n_{N_2})^2}{n_{N_2}} \right) \frac{\bar{R}T_1}{p_1 V} \quad (41)$$

Solving this quadratic equation will give a physical and non-physical solution for the post-shock condition. With the physical solution the number of moles of atomic nitrogen can be found using:

$$n_N = 2(n_{N_2} - x_{\text{physical}}) \quad (42)$$

Here x_{physical} is the physical root of the quadratic equation in Eq. (41). These new solutions become the boundary conditions of the shock wave model.

The CHEMKIN software package [14] was utilized to analyze both reacting and inert shock waves. CHEMKIN provides a comprehensive database of chemical and thermodynamic

data for molecular species, enabling the formulation and solution of complex chemical kinetics problems. Its use was essential for this study, as it accurately updates the thermodynamic properties of the species N and N₂ across the shock wave.

4 Shock Wave Structure

The computational procedures are presented below, providing the framework through which the shock wave structure is examined. Both inert and reactive flow studies are considered, each offering insight into the influence of viscous effects. In the inert case, the analysis reveals solely mass dissipation, while the reactive case highlights the thermochemical reaction that can take place due to the temperature change. Together, these approaches offer a comprehensive view of the shock structure.

4.1 Parameters

This subsection outlines the input parameters used in both the inert and reactive studies. These cases share the same initial conditions, including the initial number of moles of N and N₂, as well as the total mass and other relevant thermodynamic and transport properties. The parameters used in CHEMKIN can be seen in Table 1. For this study the Prandtl number, Pr, was chosen to be 3/4 in order to define all diffusive constants.

Table 1: Properties of N N₂ mixture from CHEMKIN

Variable	Value	Units
n_N	1.0×10^{-5}	kmol
n_{N_2}	1.0×10^{-2}	kmol
c_v	0.7411	kJ/kg/K
c_p	1.0381	kJ/kg/K
R	296.8607	kJ/kg/K
γ	1.40064	
μ	0.17899	Pa s
k	0.2477	kJ/s/m/K
$\mathcal{D}$	0.3193	m ² /s

In Table 1, the values below the dotted line represent the calculated properties given the assumption that the Prandtl number, $Pr = 3/4$. The other properties are calculated using CHEMKIN subroutines and the initial temperature.

4.2 Rankine–Hugoniot Jump Conditions

By defining the initial shock parameters, the final thermodynamic properties of the system are calculated using the Rankine-Hugoniot jump conditions. These jump conditions provide a set of equations that relate the pre-shock and post-shock states across a shock discontinuity, ensuring the conservation of mass, momentum, and energy. While the system under exhibits viscosity and structural characteristics, the Rankine-Hugoniot relations remain a valid tool for analysis due to their basis in conservation laws. The presence of viscosity in the atomic nitrogen N and molecular nitrogen N_2 system does not invalidate the use of the jump conditions; instead, it only influences the shock structure. Viscosity affects how the shock dissipates energy and how the flow behaves behind the shock, but it does not alter the fundamental relationship between the pre-shock and post-shock states that the Rankine-Hugoniot conditions describe.

A fundamental assumption underlying the remainder of this report must be addressed. The current mixture was initially treated as a calorically imperfect ideal gas (CIIG); however, due to the derivation of the Rankine–Hugoniot conditions, the boundary conditions were inadvertently treated under the assumption of a calorically perfect gas (CPG). To correct this inconsistency, the subsequent derivation of the Rankine–Hugoniot relations would proceed in similar steps, but the third governing equation should adopt a more generalized form of the energy equation, appropriate for a calorically imperfect gas.

The jump relations, assuming CPG, are derived by manipulating the general simplified statements of conservation of number (mass), momentum, and energy. [15]

$$\rho_1 u_1 = \rho_2 u_2, \quad (43)$$

$$p_1 + \rho_1 u_1^2 = p_2 + \rho_2 u_2^2, \quad (44)$$

$$\frac{p_1}{\rho_1} \frac{\gamma}{\gamma - 1} + \frac{1}{2} \rho_1 u_1^2 = \frac{p_2}{\rho_2} \frac{\gamma}{\gamma - 1} + \frac{1}{2} \rho_2 u_2^2. \quad (45)$$

Using these three conservation equations, we can calculate the post shock conditions, using the defined pre-shock state. A shock wave flows through within the wave reference frame, the unshocked material is moving at speed u_1 .

From the mass and linear momentum,

$$\rho_1 u_1 (u_1^2 - u_2^2) = (p_2 - p_1). \quad (46)$$

We can further simplify using mass conservation which gives,

$$u_1^2 - u_2^2 = \frac{p_2}{\rho_2 u_2} - \frac{p_1}{\rho_1 u_1} \quad (47)$$

p_1/ρ_1 and p_2/ρ_2 can be eliminated by employing the relationships from Eq. (45). The pressure ratio can further be derived from Eq. (47), which concludes,

$$\frac{p_2}{p_1} = \frac{2\gamma}{\gamma + 1} M_1^2 - \frac{\gamma - 1}{\gamma + 1} \quad (48)$$

Here M_1 is the Mach number at the pre-shock condition, initially set to 2.5. Then using Eq. (45), the density ratio can be simplified to the following,

$$\frac{\frac{\gamma + 1}{\gamma - 1} + \frac{p_1}{p_2}}{1 + \frac{\gamma + 1}{\gamma - 1} \frac{p_1}{p_2}} \quad (49)$$

Finally, the temperature ratio across the shock wave is determined directly from the equation of state.

$$\frac{T_2}{T_1} = \frac{\rho_1 p_2}{\rho_2 p_1} \quad (50)$$

These Rankine-Hugoniot jump conditions can be visualized across a varying initial mach number in Figure 3.

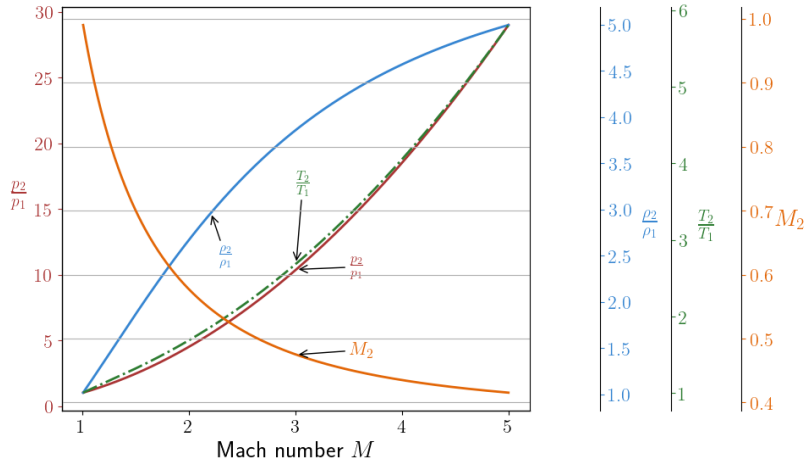


Figure 3: Rankine-Hugoniot jump relations

This model gives a close representation of the equilibrium points of the system, but, the normal Rankine-Hugoniot jump conditions do not perfectly describe mixture. To understand the effect that the reaction has on the jump conditions, they are modeled using the ratio of specific heats across the total shockwave.

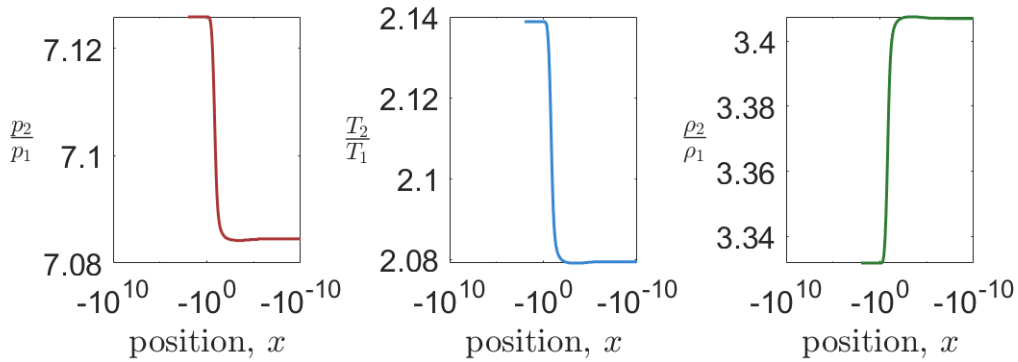


Figure 4: Rankine-Hugoniot jump relations across shock wave

Figure 4 illustrates the effect of the reaction on the Rankine-Hugoniot jump conditions.

The effect of the nitrogen dissociation within the shock wave induces a 2.8% decrease of the Temperature ratio across the shock, where as the density and pressure ratios were decreased by 1.7% and 0.5%. To simplify the boundary conditions of this study, due to the small effect of the reaction on the thermodynamic ratios across the shockwave, the jump conditions will be defined using the initial thermochemical properties of the system. As such, the boundary conditions of the N and N₂ are defined in Table 2.

Table 2: Boundary conditions of shock properties

Equilibria	Variable	Value	Units
pre-shock	M_1	2.5	
	u_1	882.959	m/s
	T_1	300	K
	ρ_1	5.606×10^{-1}	kg/m ³
	p_1	49.921	kPa
post-shock	M_2	0.513	
	u_2	265.186	m/s
	T_2	642.0	K
	ρ_2	1.867	kg/m ³
	p_2	355.749	kPa

Here the subscript X₁ or X₂ denotes the position of that variable. 1 corresponds to the pre-shock condition of the system, whereas 2 is defined as the post-shock condition.

4.3 Inert Shock Computations

For the simplified limits, the reaction rate, $\dot{\omega}$, equals zero and the species properties are taken to be the same. The shock takes place under steady-state conditions, and the governing equations for the viscous shock with diffusion and no reaction are as follows. [16]

$$u \frac{d\rho}{dx} + \rho \frac{du}{dx} = 0, \quad (51)$$

$$\rho u \frac{du}{dx} + \frac{d}{dx}(p - \tau) = 0, \quad (52)$$

$$\frac{d}{dx} \left(\rho u \left(h + \frac{u^2}{2} \right) + \tau u - q \right) = 0, \quad (53)$$

$$\frac{d}{dx} (\rho Y_i u + j_i) = 0, \quad (54)$$

$$q = -k \frac{dT}{dx} + \sum_{i=1}^2 j_i h_i, \quad (55)$$

$$j_i = -\rho \mathcal{D} \frac{dY_i}{dx}, \quad (56)$$

$$\tau = \frac{4}{3} \mu \frac{du}{dx}, \quad (57)$$

$$c_p = Y_N c_{p_N} + Y_{N_2} c_{p_{N_2}}, \quad h = c_N h_N + c_{N_2} h_{N_2}, \quad (58)$$

$$c_v = Y_N c_{v_N} + Y_{N_2} c_{v_{N_2}}, \quad e = c_{p_N} e_N + c_{p_{N_2}} e_{N_2}, \quad (59)$$

$$p_i = \rho_i R_i T \quad (60)$$

4.3.1 System of Equations

In its current form, the system contains 10 unknowns, with enough equations to solve for each. However, simplifying the system using algebraic constraints reduces both the computational time and energy required. This not only improves efficiency but also makes the solver more robust for extended simulations. Starting with linear momenta, Eq. (81) the first differential equation can be solved for du/dx

$$\rho u \frac{du}{dx} + \frac{d}{dx}(p - \tau) = 0, \quad (61)$$

$$\frac{d}{dx} (\rho u^2 + p - \tau) = 0. \quad (62)$$

After applying the product rule, the conservative form of the linear momenta equation remains. τ can then be substituted into the equation to achieve:

$$\frac{d}{dx} \left(\rho u^2 + p - \frac{4}{3} \mu \frac{du}{dx} \right) = 0. \quad (63)$$

Once in the conservative form, the differential equation can be integrated and the boundary conditions can be applied as $x \rightarrow \infty$: which yields

$$\rho_1 u_1^2 + p_2 - \underbrace{\frac{4}{3} \mu \frac{du}{dx}}_{=0} = C_1, \quad (64)$$

$$\frac{d}{dx} \left(\rho u^2 + p - \frac{4}{3} \mu \frac{du}{dx} \right) = \rho_1 u_1^2 + p_2. \quad (65)$$

After slight algebraic rearranging, Eq. (65) becomes:

$$\frac{du}{dx} = \frac{3}{4\mu} (\rho_1 u_1^2 + p_1 - (\rho u^2 + p)). \quad (66)$$

The temperature gradient can be derived from the energy equation.

$$\frac{d}{dx} \left(\rho u \left(h + \frac{u^2}{2} \right) + \tau u + k \frac{dT}{dx} + \sum_{i=2}^2 j_i h_i \right) = 0. \quad (67)$$

At this stage, we can integrate the differential equation and apply the boundary conditions as $x \rightarrow \infty$,

$$\left(\rho_1 u_1 \left(h_1 + \frac{u_1^2}{2} \right) + \underbrace{\tau_1}_{=0} u_1 + \underbrace{k \frac{dT}{dx} + j_{N_1} h_{N_1} + j_{N_{2,1}} h_{N_{2,1}}}_{=0} \right) = C_2 \quad (68)$$

Here τ_1 is the shear stress in the far field which is zero. Using the boundary conditions set from the linear momenta equation in Eq. (65), τ can be solved in terms of the pressure,

density and velocity,

$$\tau = \rho u^2 + p - (\rho_1 u_1^2 + p_1) \quad (69)$$

After applying boundary conditions, and substituting Eq. (69), the temperature gradient becomes,

$$\frac{dT}{dx} = \frac{\rho_1 u_1 (h_1 + u_1^2/2) - \rho u (h + u^2/2) + (\rho_1 u_1^2 + p_1 - (\rho u^2 + p)) u + j_N h_N + j_{N_2} h_{N_2}}{k}. \quad (70)$$

Applying the mass diffusion constraint, Eq. (73) simplifies to,

$$j_N + j_{N_2} = 0, \quad (71)$$

$$j_N h_N + j_{N_2} h_{N_2} = j_N (h_N - h_{N_2}), \quad (72)$$

$$\frac{dT}{dx} = \frac{\rho_1 u_1 (h_1 + u_1^2/2) - \rho u (h + u^2/2) + (\rho_1 u_1^2 + p_1 - (\rho u^2 + p)) u + j_N (h_N - h_{N_2})}{k}. \quad (73)$$

Because the initial computation examines the inert shock structure of the N and N₂, the molar production rate, $\dot{\omega}$, vanishes from the species evolution equation. This simplification eliminates chemical source terms and allows the analysis to focus solely on diffusive and convective transport. Beginning with the species evolution equation and applying Fick's law Eq. (85), we obtain:

$$\frac{d}{dx} \left(\rho Y_i u - \rho \mathcal{D} \frac{dY_i}{dx} \right) = 0. \quad (74)$$

Integrating this equation and applying the appropriate boundary conditions yields an expression for the spatial rate of change of species concentrations, driven purely by flow and diffusion.

$$\frac{dY_i}{dx} = \frac{(\rho Y_i u - \rho_1 Y_{i,1} u_1)}{\rho \mathcal{D}} \quad (75)$$

Here the subscripts attached to the concentration, $Y_{i,1}$, show the different species and that it is the boundary condition of the chosen species. With Eq. (75), the system is fully defined. The differential equations which are numerically solved are restated below:

$$\frac{du}{dx} = \frac{3}{4\mu} (\rho_1 u_1^2 + p_1 - (\rho u^2 + p)), \quad (76)$$

$$\frac{dT}{dx} = \frac{\rho_1 u_1 (h_1 + u_1^2/2) - \rho u (h + u^2/2) + (\rho_1 u_1^2 + p_1 - (\rho u^2 + p)) u + j_N (h_N - h_{N_2})}{k}, \quad (77)$$

$$\frac{dY_N}{dx} = \frac{(\rho Y_N u - \rho_1 Y_{N,1} u_1)}{\rho \mathcal{D}}, \quad (78)$$

$$\frac{dY_{N_2}}{dx} = \frac{(\rho Y_{N_2} u - \rho_1 Y_{N_2,1} u_1)}{\rho \mathcal{D}}. \quad (79)$$

4.3.2 Results and Discussion

Using the methods outlined in Section 3, a Fortran script was written to solve the system of differential equations. To converge on a solution, the boundary conditions were slightly perturbed off of the post shock solution then integrated backwards along the shock front. Figure 5 shows the results of the direct integration.

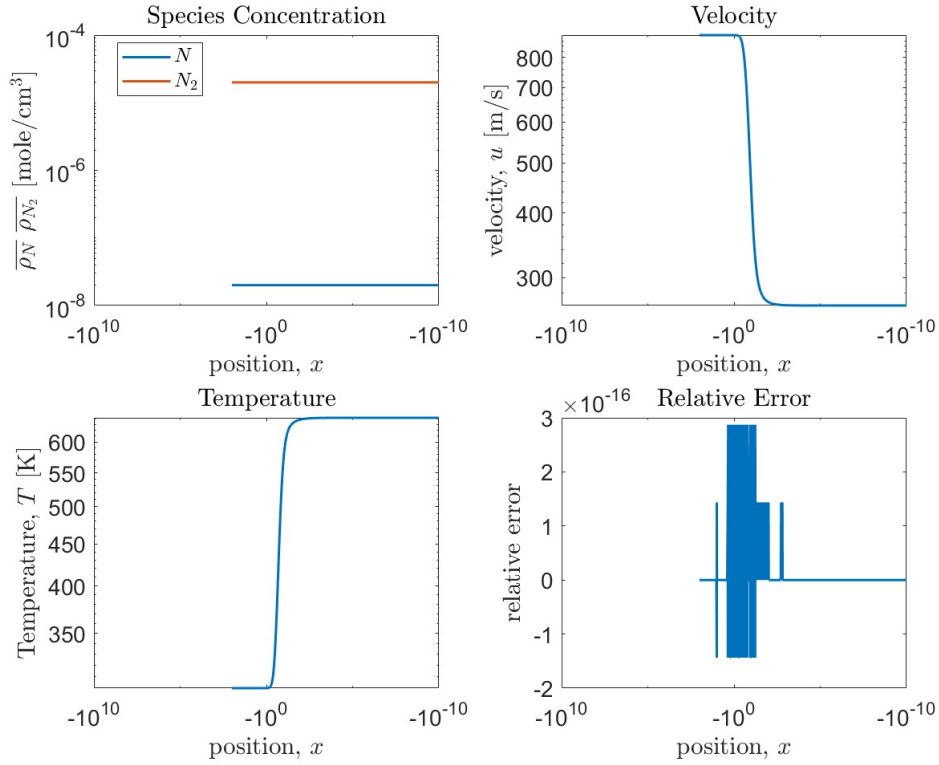


Figure 5: Results of inert shock wave analysis

In all subsequent figures, the pre-shock state is presented on the left and the post-shock state on the right. The upper-left panel displays the species concentrations across the shock front. As expected under the inert assumption, the initial concentrations of atomic nitrogen N and molecular nitrogen N_2 remain effectively constant across the shock. This behavior is consistent with the absence of chemical reactions in the inert N and N_2 , resulting in no species evolution throughout the shock structure.

The velocity profile shows a clear transition from supersonic conditions upstream of the shock to subsonic conditions downstream, aligning with the physical behavior of a normal shock in compressible flow. The integration results also converge to the results of the analysis done in Section 4.2.

The temperature profile, shown in the lower-left panel, highlights the sharp thermal gradient across the shock front, illustrating both the magnitude of the temperature rise and the finite shock thickness. Following the numerical integration, the temperature smoothly

decays back to the ambient pre-shock value of 300 K, consistent with upstream boundary conditions. This convergence confirms the stability of the integration methods.

Finally, the relative error of due to mass conservation was calculated to ensure the system properly obeyed the conservation laws. The relative error was osculating around a magnitude of 10^{-16} proving the mass is conserved along the shockwave.

Using the results of the numerical integration, the total viscous shear stress across the shock, and the total entropy production could be plotted.

The viscous shear was plotted by using Eq. (69) with the results of the velocity gradient. Whereas the entropy production was calculated as outlined in the CHEMKIN Manual. Figure 6 shows displays both the viscous shear stress and the entropy across the shock wave.

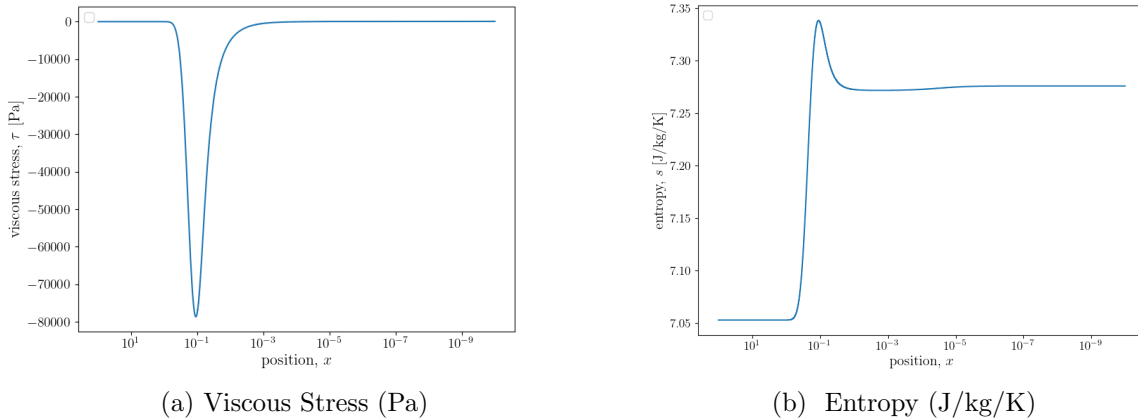


Figure 6: Viscous stress and entropy across shock

The viscous stress in the shock wave is directly tied to the shock thickness. If the shock were treated as an infinitesimally thin discontinuity, as in an inviscid model, the stress would theoretically approaches infinity. Instead, the finite thickness introduced by viscosity spreads the stress out across the shock front, lowering its overall magnitude. Additionally, the stress must return to zero in both the pre-shock and post-shock regions, since we are not accounting for turbulent stresses at the boundaries.

Next, the entropy production across the shock wave is visible in the right panel of Figure 6. As expected, entropy increases through the shock due to the rise in both temperature and

density. After the shock front, the entropy decreases slightly, indicating that a portion of the entropy produced during the shock process was reversible. What remains at the post-shock state is the irreversible contribution to the total entropy.

Next, the mass diffusion of molecular and atomic nitrogen in the mixture were calculated. Using Fick's law as outlined in Eq. (85) the total mass diffusion for both species can be found in Figure 7.

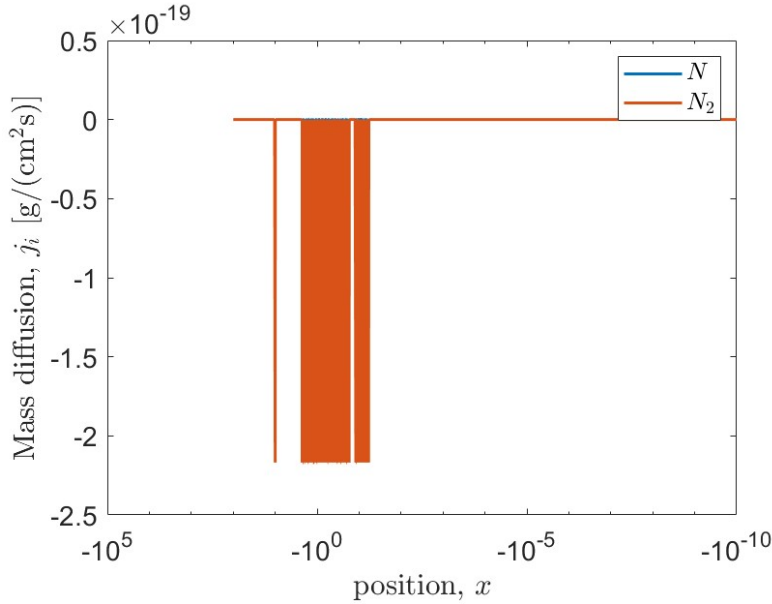


Figure 7: Mass diffusion of N- N_2 mixture across shock

Because there was no reaction, $dY_i/dx \approx 0$. We see that Y_i should remain near constant by analyzing Eq. (78) and Eq. (79). The numerator of these equations is nearly identical to the simplified expression of conservation of mass, and after the first iteration of the numerical analysis, Y_i will be forced to be equal to $Y_{i,1}$ in order to satisfy the equation.

4.4 Shock Computations with Reaction

Having established the structure of the inert atomic and molecular nitrogen shock, we now re-introduce chemical reaction to examine its influence on the shock dynamics and thermodynamic properties. For this study we will define the same N and N_2 mixture, but the

chemical evolution equation is no longer homogeneous. The governing equations for the reacting shock can be found below.

$$u \frac{d\rho}{dx} + \rho \frac{du}{dx} = 0, \quad (80)$$

$$\rho u \frac{du}{dx} + \frac{dp}{dx} - \frac{d\tau}{dx} = 0, \quad (81)$$

$$\rho u \frac{de}{dx} + \frac{dq}{dx} + (p - \tau) + \frac{du}{dx} = 0, \quad (82)$$

$$\rho u + \frac{dY_i}{dx} + \frac{dj_i}{dx} = M_i \dot{\omega}_i, \quad (83)$$

$$q = -k \frac{dT}{dx} + \sum_{i=1}^2 j_i h_i, \quad (84)$$

$$j_i = -\rho \mathcal{D} \frac{dY_i}{dx}, \quad (85)$$

$$\tau = \frac{4}{3} \mu \frac{du}{dx}, \quad (86)$$

$$c_p = Y_N c_{pN} + Y_{N_2} c_{pN_2}, \quad h = c_N h_N + c_{N_2} h_{N_2}, \quad (87)$$

$$c_v = Y_N c_{vN} + Y_{N_2} c_{vN_2}, \quad e = c_{pN} e_N + c_{pN_2} e_{N_2}, \quad (88)$$

$$p = \rho RT \quad (89)$$

4.4.1 System of Equations

In order to solve the reactive model a similar system needs to be constructed utilizing the governing equations. The difference between the two studies lies within the homogeneity of the chemical evolution equation, as such, we can no longer simply integrate and apply the boundary conditions as before. Therefore, matrix solution is one approach to single out each differential equation from the governing and constitutive models. Variables left out of the matrix model are utilized as supplementary, and calculated using algebra.

We can set up the matrix form of the governing equations as such:

$$\underline{A} = \begin{bmatrix} \rho u & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & \rho u & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & \rho u - \frac{\rho}{u} RT & -1 & 0 & 0 & \rho R \\ 0 & 0 & p - \tau & 0 & 1 & 0 & c_v \\ -\rho \mathcal{D} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{4}{3\mu} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -k \end{bmatrix} \quad \vec{b} = \begin{bmatrix} M_N \dot{\omega}_N \\ M_{N_2} \dot{\omega}_{N_2} \\ 0 \\ 0 \\ j_N \\ \tau \\ q - j_N (h_N - h_{N_2}) \end{bmatrix} \quad (90)$$

By inverting the matrix we are able to isolate each of the differential equations. We define the dynamical system:

$$\vec{Z} = \begin{bmatrix} Y_N \\ N_2 \\ u \\ \tau \\ q \\ j_N \\ T \end{bmatrix} \quad (91)$$

Once the system of state variables are defined, the matrix and constants defined in Eq. (90) are inverted to solve for the system of equations.

$$\underline{A} \frac{d\vec{Z}}{dx} = \vec{b} \quad (92)$$

$$\frac{d\vec{Z}}{dx} = \underline{A}^{-1} \vec{b} \quad (93)$$

Once defining the coefficient matrix, $\underline{A}$, the coefficient vector, $\vec{b}$ and the solution vector,

$\frac{d\vec{Z}}{dx}$ is defined:

$$\frac{d\vec{Z}}{dx} = \begin{bmatrix} \frac{-j_N}{\mathcal{D}\rho} \\ \frac{uj_N + \mathcal{D}M_N\dot{\omega}_N + \mathcal{D}M_{N_2}\dot{\omega}_{N_2}}{\mathcal{D}u\rho} \\ \frac{3\mu\tau}{4} \\ \frac{-4qRu\rho - 3kRT\mu\rho\tau + 3ku^2\mu\rho\tau + 4Ru\rho h_N j_N - 4Ru\rho h_{N_2} j_N}{4ku} \\ \frac{-3kp\mu\tau + 3k\mu\tau^2 + 4qc_v - 4c_v h_N j_N + 4c_v h_{N_2} j_N}{4k} \\ \frac{uj_N + \mathcal{D}M_N\dot{\omega}_N}{\mathcal{D}} \\ \frac{-q + h_N j_N - h_{N_2} j_N}{k} \end{bmatrix} \quad (94)$$

these equations can then be implemented in the Fortran code to model the reaction of the species. Once all of the equations are isolated, the thermodynamic properties were then calculated across the shock wave.

4.4.2 Results and Discussion

Using the methods outlined in Section 3, the isolated system of differential equations were used to analyze the shock wave. Similarly to the inert model, to converge to a solution, the boundary conditions were slightly perturbed off of the post shock conditions then integrated backwards along the shock front. Figure 8 shows the results of direct integration.

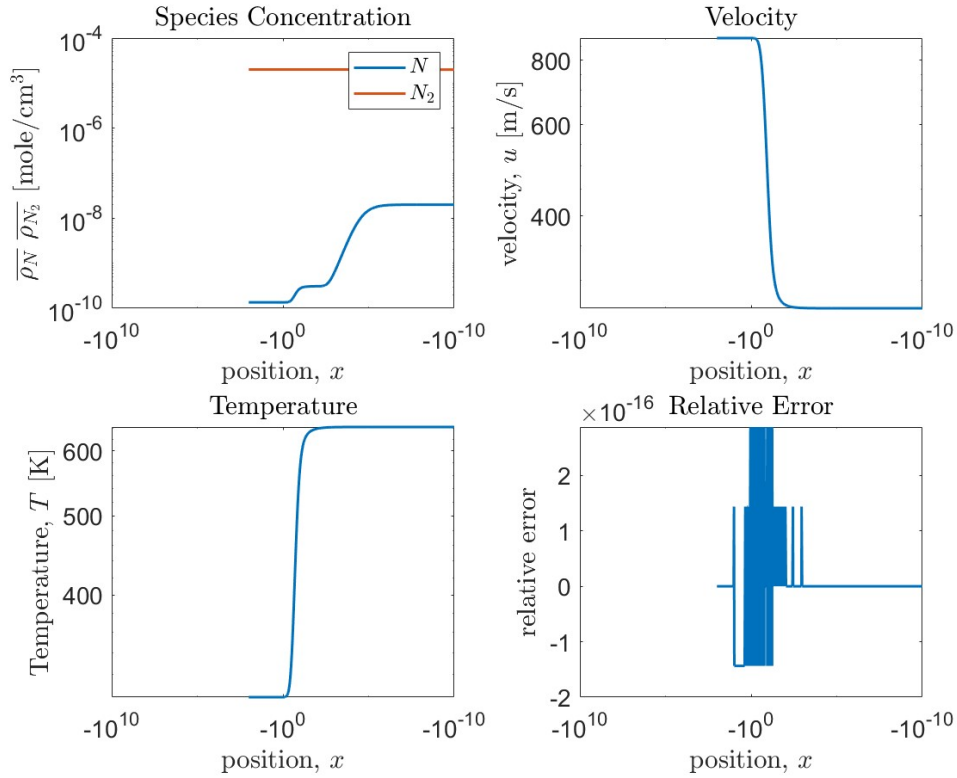


Figure 8: Direct numerical integration of reactive shock front

The top-left panel illustrates the concentration profiles of atomic and molecular nitrogen across the shock structure. The atomic nitrogen concentration rises slightly after the shock front, followed by a more pronounced increase farther downstream. This delayed rise resembles the behavior seen in square-wave detonations, where the induction zone, the region between the shock and the subsequent exothermic reaction—precedes the main chemical event. In our case, the concentration rises again after the initial shock, suggesting an endothermic reaction where the nitrogen dissociation is an endothermic reaction. While the accompanying temperature and species plots do not indicate a second detonation, the observed behavior suggests that the dissociation of N_2 might exhibit a thermally activated delay in response to the elevated post-shock temperature.

The temperature and velocity profiles both asymptotically converge to their post-shock values predicted from Section 4.2. This convergence is important because the implementation

of the reaction should not interfere with the flow parameters.

Lastly, the relative mass error, shown in the bottom right panel of Figure 8, oscillates at a magnitude of 10^{-16} . This behavior is consistent with the inert shock wave. This indicates that the solver maintains numerical stability and mass conservation, even in the presence of weak chemical activity.

Similarly to the inert solution, the viscous stress and entropy production were plotted along the shock wave. The results of their calculations can be found in Figure 9

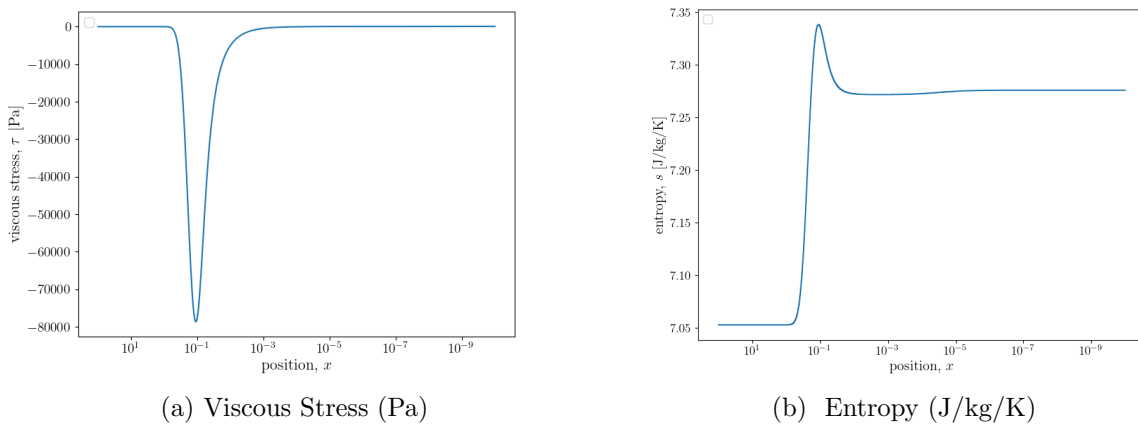


Figure 9: Viscous stress and entropy across shock

Compared to the inert shock wave, the reaction does not appear to have substantial impact on the stress or entropy production. This is due to the low temperature change across the shock, because the reaction is not prominent enough to have a noticeable effect on the properties of the mixture.

The reaction itself is not possible without the molar production rate. Using Arrhenius kinetics the molar production rate was calculated and plotted across the shock wave. Figure 10 shows how the production rate changed over the course of the shock wave.

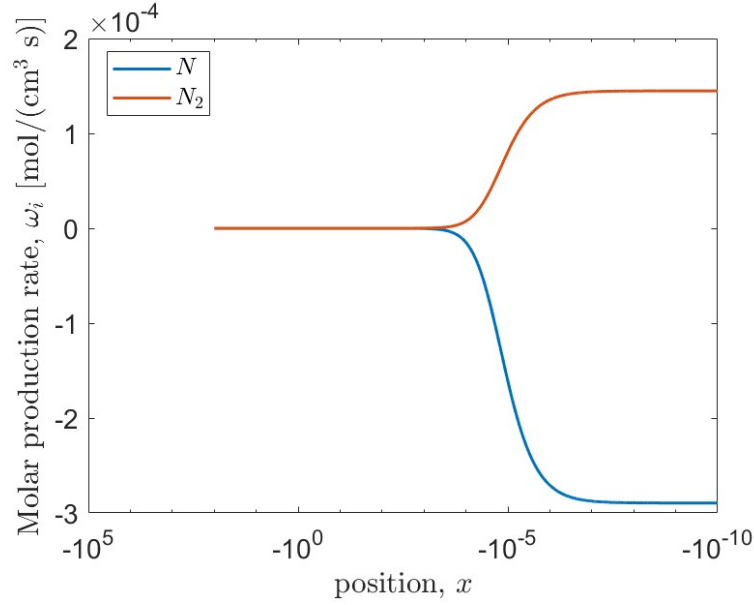


Figure 10: Molar production rate along the shock wave

At first glance, the figure may appear counterintuitive due to the presence of a nonzero molar production rate at the post-shock equilibrium. However, this can be attributed to the nature of the numerical integration. In order to integrate smoothly along the shock profile, the boundary condition was applied slightly within the post-shock region rather than at the true equilibrium state. As a result, the system has not fully equilibrated at that point, and a finite rate of production is expected. Notably, the rate also reflects a 2:1 ratio of atomic to molecular nitrogen formation. Finally, the mass diffusion across the reacting shock wave is shown in Figure 11

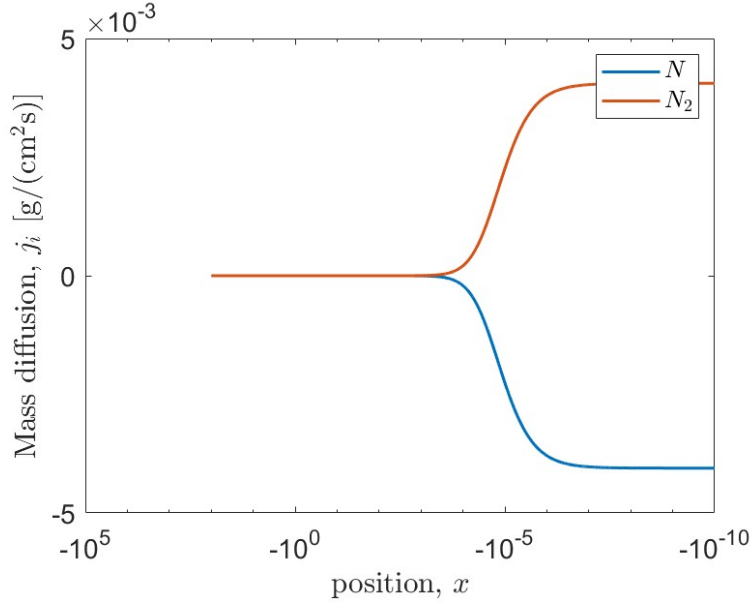


Figure 11: Mass diffusion of molecular and atomic nitrogen within a shock.

The mass diffusion across the shock wave obeys the constraint we put on it in Eq. (71), $\sum j_i = 0$. Ultimately, the differences between the reactive and inert shock structures may be indistinguishable to the naked eye, but when dealing with areas where these time and length-scales matter, it becomes essential to account for chemical reactions and diffusive transport.

4.4.3 Effect of Shock Strength on Molecular Dissociation

By increasing the initial flow velocity, the shock structure can be analyzed across the transonic and supersonic regions. Increasing the initial velocity will give a better understanding of the effects of viscosity in these high speed regions. The following calculations were done using the procedure outlined in Section 3, Chemical Kinetics. Each iteration needed to be slightly perturbed off of their post-shock equilibrium point in order to integrate the system from the two states of equilibrium. The post-shock thermodynamic properties were calculated by calculating the nonlinear Rankine-Hugoniot space curves. By varying the initial Mach number, for the reactive study, we can see how the stronger shock effects the reaction and flow properties. Figure 12 shows the velocity gradient across the shockwave

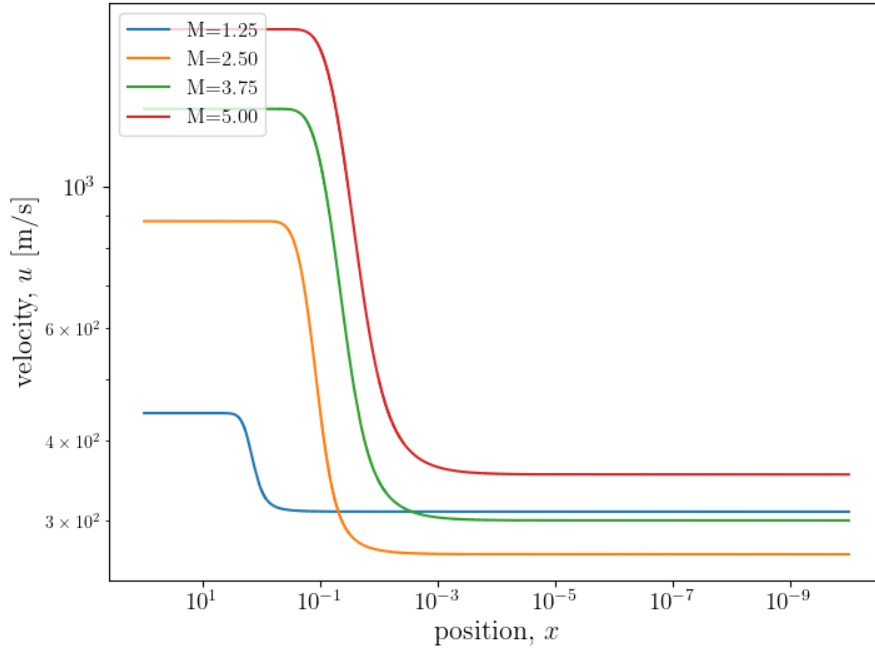


Figure 12: Velocity gradient

Figure 12 presents a comparison of the velocity profiles of shock waves corresponding to different initial Mach numbers. While each profile exhibits the characteristic viscous shock structure, this structure remains qualitatively similar across varying Mach numbers. However, the post-shock velocities do not converge to the same value. These discrepancies arise from differences in the Rankine–Hugoniot jump conditions applied for each case. Notably, the post-shock states in all cases approach their respective inviscid projections and remain subsonic, as expected.

Although the overall structure of the shock profiles may appear similar, subtle variations occur as a function of Mach number. To approximate these differences the shock thickness can be used to calculate the shock length. The mean free path is estimated:

$$\lambda \approx \frac{7.8895 \times 10^{-9} \frac{\text{g}}{\text{cm}^2}}{\rho_1 \frac{\text{g}}{\text{cm}^3}}, \quad (95)$$

where λ represents the mean free path between particles in the fluid. The corresponding

shock thickness can then be estimated using the following expression:

$$l_{\text{shock}} \approx \frac{8\gamma\mu}{3(\gamma+1)\rho_1(u_1-u_2)} \sqrt{\frac{RT}{3\mathcal{M}}}. \quad (96)$$

Here, l_{shock} denotes the characteristic length scale of the shock. The resulting values for shock thickness at various Mach numbers are summarized in Table 3.

Table 3: Shock thickness and mean free path for N N₂ mixture

Mach Number	Shock Thickness (mm)	Mean Free Path (mm)
1.25	2.8494×10^{-4}	1.4075×10^{-4}
2.5	6.1060×10^{-5}	1.4075×10^{-4}
3.75	3.6812×10^{-5}	1.4075×10^{-4}
5.0	2.6714×10^{-5}	1.4075×10^{-4}

Table 3 is used to describe the differences in shock structure for the varying mach number. As the mach number increases the shock thickness decreases. This is due to the strength of the shock increasing, leading to a thinner shock layer.

5 Conclusions

The structure of a viscous shock wave was determined through a fully analytical approach. By incorporating chemical kinetics, the evolution of temperature across the shock wave was used to predict the extent of molecular nitrogen dissociation. Solving the governing conservation equations ensured continuity across the shock structure, thereby enabling the resolution of viscous effects on the shock profile. The numerical solution demonstrated excellent mass conservation, with the total relative error on the order of 10^{-16} . Additionally, analysis of varying Mach numbers revealed that the shock thickness decreases significantly with increasing Mach number, from 6.106×10^{-5} mm at $M = 2.5$ to 2.671×10^{-5} mm at $M = 5.0$.

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